

[Total No. of Questions : 12] [Pages : 2]

DEC-25-0507
MCA-7306(v) (Software Engineering)
MCA-3rd NEP

Time : 3 Hours

Max. Marks : 60

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note: Attempt five questions in all, select one question each from section A, B, C, D. Section E (Question-9) is compulsory.

Section A

1. Discuss software engineering myths in detail. How do they impact project success? (12)

2. Define the term Software Development Life Cycle (SDLC). Explain in detail the different phases of the Waterfall Model, illustrating your answer with a neat and well-labeled diagram. (12)

Section B

3. (a) With the help of suitable equations, explain how effort, development time, and staffing are estimated using Basic COCOMO. (8)

- (b) State the limitations of the COCOMO model. (4)

4. What is Agile Methodology? Discuss how Agile ensures better risk management compared to traditional SDLC models. (12)

Section C

5. (a) Discuss cohesion and coupling in function-oriented design. Why are they important for modular design? (6)

2

MCA-7306(v)

- (b) Write short note on CRC cards. (6)

6. Explain the significance of Unified Modeling Language (UML) in software development. Using a suitable case study of a Library Management System, design and illustrate the following UML diagrams: (a) Use-Case Diagram, (b) Class Diagram, and (c) Sequence Diagram. (12)

Section D

7. (a) A software project had a development cost of ₹20,00,000. (6)

If the annual maintenance cost is 60% of the development cost, calculate:

- The annual maintenance cost.
- The total cost of ownership (TCO) after 5 years.

- (b) Compare token count metrics and data structure metrics with suitable examples. (6)

8. Define Web Engineering Project Metrics. Discuss their role in improving quality and productivity of web-based systems. (12)

Section E (Compulsory)

9. (a) State any two differences between product and process. (2)

- (b) Define a "module" in function-oriented design. (2)

- (c) List two advantages of having a well-defined software architecture. (2)

- (d) What is meant by a "user story" in Agile? (2)

- (e) When is regression testing performed? (2)

- (f) Differentiate between class and object. (2)